INCH-POUND
MIL-PRF-81757/21(CR)
W/AMENDMENT 1
9 MAY 2016
SUPERSEDING
MIL-PRF-81757/21(CR)
25 FEB 2011

PERFORMANCE SPECIFICATION SHEET

BATTERY, RECHARGEABLE, REPAIRABLE, NICKEL CADMIUM, VENTED, 24-VOLT, 13 AMPERE-HOUR, BB-664/A

This specification is approved for use within Army CECOM Life Cycle Management Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product herein shall consist of this specification and MIL-PRF-81757.

REQUIREMENTS

1. Part or identifying number. The part or identifying numbers (PIN) for the items covered by this specification sheet are as follows:

PIN	Description	Figures
M81757/21-1	BB-664/A Battery	1-4
M81757/21-2	Connector, Strap (common link)	5
M81757/21-3	Connector, Strap (curved link)	5

2. Dimensions and weight. The dimensions and weight of the battery (or cell) shall be as specified in 3.10.8 of MIL-PRF-81757 as modified herein and as shown on figure 1 herein. The weight of the battery shall be not greater than 27 pounds (12.2 kg).

3. Capacity. The minimum capacity of the battery shall be 13 ampere-hours (14.2 initially). When tested as specified in MIL-PRF-81757, the test parameters and requirements of Table I herein shall apply.

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TABLE I: Test parameters and requirements, BB-664/A battery

Test	Test Method 1/	Discharge Rate(A) 2/	Cut Off Voltage(V) 2/	Capacity (Ah) Minimum 2/
Initial capacity discharge	4.5.13	13	18	14.2
Humidity and Charge Retention	4.5.14	13	18	9.7
Shock	4.5.15	330	10	5.5
Vibration	4.5.16	330	10	5.5
Temperature Shock	4.5.17	13	18	13
Altitude	4.5.18	13	18	13
Twenty-second pulse discharge	4.5.20	330	10	5.5
Medium Rate Discharge and Operating Position	4.5.21	117	14.4	9.7
Cycle Life	4.5.23			
Normal Rate (Cycle 1)		13	18	13
High Rate		330	10	3.7
Normal Rate (Cycle 100)		13	18	13
Normal Rate (Cycle 200)		13	18	13

1/ Test methods are referenced from MIL-PRF-81757.
2/ Test parameters and requirements cited herein supersede those in the general specification.

4. Electrical interface requirements. The BB-664/A battery shall have two connectors, located as described by Figure 1:

a. Connector J1: a 2-contact MS3509-XX receptacle, qualified in accordance with specification MIL-PRF-18148
b. Connector J2: a 13-contact connector that meets the interface requirements of MS27505E11E35S

5. Cells. The BB-664/A shall include 19 each BB-475/A cells, PIN M81757/22-1. Cells shall be connected inside the BB-664/A battery using fifteen M81757/21-2 common links and one M81757/21-3 curved link to connect to the positive MS3509 connector.

6. Cell links. The cell links used for connecting cells in series shall be copper with a cross-sectional area not less than 0.039 in^2, with nickel plating that complies with SAE-AMS-QQ-N-290, Class 1, Grade E. Finish shall be smooth and free of corrosion, pits, and blow-holes.

7. Wiring harness. The BB-664/A shall include 1 wiring harness, PIN M81757/23-1.

8. MIL-PRF-81757 variance. The battery shall comply with MIL-PRF-81757 except as follows.

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8.1 Modify the following paragraphs:

3.6.2 Fasteners. After the phrase "…free of any coating…" add the following: "except as otherwise specified herein."

3.7.2 Battery container and cover.

a. Delete the fifth sentence ("The cover shall be fitted…the container body."). The BB-664/A requires a gap for ventilation (See Figure 2 and Figure Note 4 herein).

b. Add the following: "When tested as specified in 4.5.10, the battery shall show no signs of KOH leakage or seepage after testing."

3.7.4 Latches. This requirement is not applicable to the BB-664/A battery.

3.7.5 Handles. This requirement is not applicable to the BB-664/A battery.

3.7.9 Connectors, intercell. This battery is designed for full maintenance; tamper-resistant fasteners shall not be used.

3.10.22 Constant voltage discharge. This requirement is not applicable to this battery.

8.2 Add the following new paragraph:

"3.7.11.9 Shipping condition. All batteries shall be delivered complete with electrolyte to the proper level and completely discharged, but fully formed. Each battery shall be equipped with a shorting device across the output terminals."

"3.8.4.3.1 Warning marking. To be included in the marking described by figures 2 and 3, the following marking shall appear on each battery:

WARNING

1. Area around battery must be well ventilated and cover removed during charge.

2. Do not allow caustic electrolyte to contact body, clothing, tools, etc. If contact is made wash immediately with water. Flush eyes with water only for 15 minutes and seek medical attention.

CAUTION
DO NOT USE ACID
ELECTROLYTE

Use 0.17 ± 0.030 inch lettering for this marking

Acid will permanently damage this battery.
Avoid using tools which have been used on
acid type storage batteries. Consult technical
manual before servicing.

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Use 0.09 ± 0.030 lettering for this marking

INSTRUCTION

1. Remove cover and brush off white carbonate deposits with vents in place.

2. Remove shorting plug or spring.

3. Refer to charging instructions in applicable technical manuals.

4. After charge, allow 1-2 hour stand, remove cell vents and adjust electrolyte using distilled water to 1/4 inches above plates for the cells. Replace cell vents and cover.

5. Periodically soak cell vents in distilled water.

Use 0.09 ± 0.030 lettering for this marking"

"3.8.8 Warning Tags. Where shipping devices are furnished by the contractor, alkali resistant tags with strong alkali resistant nonmetallic cord shall indicate action or actions to be taken by field personnel."

"3.10.31 Testing of wiring harness. When the wiring harness is subjected to either the wiring harness temperature test or the electrolyte resistance test, the wiring harness shall exhibit the following:

"a. Continuity. When tested as specified in 4.5.32, continuity results shall be as follows:

(1) Test points of 4.5.32a: Continuity

(2) Test points of 4.5.32b: Resistance (see paragraph 2f, MIL-PRF-81757/23)

(3) Test points of 4.5.32c: Not less than 1.0 megohm.

"b. There shall be no evidence of alkali migration into harness wires and components; breakdown of insulation; stripping of metal plating from connectors; corrosion of metal parts; nor cracking of potting compound.

"c. The transducer shall meet the requirements herein.

"d. The thermoswitches shall close or open within the specified temperature ranges herein.

"e. There shall be no mechanical failure of components.

"f. There shall be no dimensional distortion beyond specified limits."

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"3.10.32 Heater blanket. The M81757/21-1 (BB-664/A) battery shall have a heater blanket consisting of three heater as strips. It shall have a total power of 342 ± 15 watts of heat uniformly delivered to the battery cells. When tested as specified in 4.5.34 installed in a M81757/21-1 (BB-664/A) battery, it shall be capable of raising the cell temperature from -54°C (-65°F) to -10°C (+14°F) in not less than one hour after the start of heating. Heater hot spot temperature shall not exceed 110°C (230°F). When tested as specified in 4.5.35, the heater shall demonstrate operation for 500 hours without deterioration or breakdown under visual examination or dielectric strength testing or change of resistance."

VERIFICATION

1. MIL-PRF-81757 variance. The battery shall comply with MIL-PRF-81757 except as follows.

1.1 Modify the following paragraphs:

4.2.1 Inspection of batteries.

a. Delete "tamper-resistant fasteners (see 3.7.9)" and replace with "two #8-32 UNC 2A screws and two M81757/1-8A spring tension washers"

b. Add: "In addition, one wiring harness, PIN M81757/23-1, shall be provided."

4.5.3 Electrolyte resistance. Add the following at the end of the paragraph: "Functional testing shall consist of the connection continuity of harness test of MIL-PRF-81757/23."

4.5.9 Handle strength test. Not applicable to this battery.

4.5.10 Cover seal test. Delete in its entirety and replace with the following: "The following test shall be performed for the BB-664/A. Place an empty battery container on a clean dry surface and fill with 1.30 ± 0.04 specific gravity KOH to within ½" of the J1 connector opening. Allow to stand for not less than 24 hours. Battery containers shall meet the requirements of 3.7.2 as modified herein."

4.5.12 Cell vent test. Add the following after the title: "This test shall be performed on all cells in a battery. Charge, immersion, and discharge steps shall be applied to single cells whenever testing cells separately from batteries."

4.5.15 Shock test (basic design). Delete "Discharge the battery at the 9C-rate for 5 minutes. Measure the voltage of the battery at 5 minutes after the start of the discharge." Replace with: "Discharge the battery at a constant current of 330 amperes in three twenty-second pulses with 120 seconds of rest between pulses."

4.5.16 Vibration test. Delete text in its entirety and replace with the following:

"The following test shall be performed. Upon completion, batteries shall meet the requirements of 3.9.

"a. Remove cover from each battery.

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"b. Constant current charge each battery at the 1C rate.

"c. Secure cover to each battery.

"d. Attach batteries to vibration equipment in the normal, upright position.

"e. Connect battery output terminals to the test system.

"f. Attach no less than two accelerometers to each battery in the direction of the vibration motion; use the spectral density average of accelerometers to control test levels.

"g. Subject batteries to the vibration test of MIL-STD-810, Method 514, Procedure I (Operational service), using Annex D parameters for Category 14, Helicopters, vibration parameters for General Materiel, f_x values for Main Rotor of AH-64 (early).

"h. Disconnect batteries from vibration equipment; attach batteries to vibration equipment in a perpendicular axis.

"i. repeat steps e through h for a total of 3 mutually perpendicular axes.

"j. Each vibration period shall be not less than one hour.

"k. Discharge batteries at the 0.1C rate during each vibration period; monitor battery voltage during discharge.

"l. Upon completion of all three vibration periods, constant current charge batteries at the 1C rate.

"m. Discharge batteries with three 20-second pulses of 330 amperes with 120 seconds of rest between pulses.

"n. Examine batteries for the requirements of 3.9."

4.5.18 Altitude test. This test is required as described with the following exceptions:

1) For step b, use a test temperature of -1.1°C in lieu of -40°C
2) For step c, use a pressure that simulates 6,090 meters (19,980 ft.) of altitude in lieu of the 18,500 meters (60,700 ft.) pressure equivalent.

4.5.20 Twenty-second pulse discharge test. This test is applicable. However, for step (c) use a constant current of 330 amperes in lieu of the sloping discharge rate.

4.5.22 Constant voltage discharge test. This test is not applicable to this battery.

4.5.23 Cycling test. Delete text in its entirety and replace with the following:

"The following test shall be performed.

"a. Cycle 1. Charge at a constant current at the C rate to 1.55 ± 0.10 volts per cell, then 1.1 amperes for another three hours. Rest for one hour at open circuit. Then discharge at the normal rate specified herein to cut-off voltage.

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"b. Cycle 2. Charge at a constant current at the C rate to 1.55 ± 0.10 volts per cell, then 1.1 amperes for another three hours. Rest for one hour at open circuit. Then discharge at the high rate specified herein for not less than 20 seconds, followed by 260 seconds of rest, then discharge for another 20 seconds at the rate specified.

"c. Cycles 3-98. Charge at a constant current at the 1.5C rate to 1.55 ± 0.10 volts per cell and rest for one hour at open circuit. Then discharge at the high rate specified herein for not less than 20 seconds, followed by not less than 260 seconds of rest, then discharge for another 20 seconds at the rate specified.

"d. Cycle 99. Charge at a constant current at the 1.5C rate to 1.55 ± 0.10 volts per cell and rest for one hour at open circuit. Then discharge at the high rate specified herein for not less than 20 seconds, followed by not less than 260 seconds of rest, then discharge for another 20 seconds at the rate specified.

"e. Cycle 100. Repeat step a.

"f. Cycle 101. Repeat step b.

"g. Cycle 102 -198. Repeat step c.

"h. Cycle 199. Repeat step d.

"i. Cycle 200. Repeat step a.

"NOTE: Water may be added on the 100th cycle. The amount of water added shall be recorded. Reconditioning is not allowed during the cycle life test."

1.2 Add the following new paragraphs:

"4.5.32 Wiring harness temperature test. The following test shall be performed. After completion of testing, the wiring harnesses shall meet the requirements of 3.1 and 3.10.31.

"a. Remove wiring harness from battery

"b. Fixture the wiring harness so that an ohmmeter is connected across the terminals of the transducer and all thermoswitches. Note: Thermoswitches S3 and S3A require monitoring of dual output.

"c. Place the harness into an environmental chamber.

"d. Set the chamber to reach a temperature within a range of +3.4 ± 1.1°C (38°F); allow the test sample to stabilize for not less than 1 hour.

"e. Set the chamber to ramp down the temperature to not less than -5.0°C (23°F) at a rate not greater than 1°C per five minutes while monitoring the continuity of thermoswitches S3 and S3A continuously.

"f. Once thermoswitches S3 and S3A change status, record the temperature and stop the temperature decrease.

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"g. Set the chamber to reach a temperature within a range of 1.0 ± 1.1°C (34°F).

"h. Set the chamber to ramp up the temperature to not less than 11.2°C (52°F) at a rate not greater than 1°C per five minutes while monitoring the continuity of thermoswitches S3 and S3A continuously.

"i. Once thermoswitches S3 and S3A change status, record the temperature and proceed to the next step.

"j. Set the chamber to reach the first temperature set point in the table below. Once the set temperature has been reached, allow the test sample to stabilize for not less than 1 hour.

"k. Record resistance values for the thermistor.

"l. Repeat steps j and l for the second set point in the table below.

"m. Repeat step l until readings have been obtained at all 4 set points.

"n. Set the chamber to reach a temperature within a range of 52.8 ± 1.1°C (127°F); allow the test sample to stabilize for not less than 1 hour.

"o. Set the chamber to ramp up the temperature to not less than 60.0°C (140°F) at a rate not greater than 1°C per five minutes while monitoring the continuity of thermoswitch S1 continuously.

"p. Once thermoswitch S1 closes, record the temperature and stop the temperature increase.

"q. Set the chamber to reach a temperature within a range of 53.2 ± 1.1°C (127°F).

"r. Set the chamber to ramp down the temperature to not greater than 46°C (115°F) at a rate not greater than 1°C per five minutes while monitoring the continuity of thermoswitch S1 continuously.

"s. Once thermoswitch S1 opens, record the temperature and proceed to the next step.

"t. Return chamber to room ambient conditions. Allow the test sample to stabilize for not less than 1 hour. Remove the test sample and examine wiring harness for the requirements of 3.10.31.

"u. Reinstall wiring harness into battery

"Temperature Set Points, BB-664/A Wiring Harness

(1) 10.0 ± 1.1°C (50°F)
(2) 21.0 ± 1.1°C (70°F)
(3) 32.0 ± 1.1°C (90°F)
(4) 43.0 ± 1.1°C (110°F)"

"4.5.33 Heater blanket operation. The heater blanket shall be tested as follows. Install the full harness assembly into a M81757/21-1 (BB-664/A) battery. Charge the battery in accordance with 4.4.3. Store the battery at -54°C (-65°F) for not less than 16 hours. Install temperature monitoring devices on all temperature switches and thermistors, and in not less than 3 non-adjacent cells to monitor electrolyte temperature. Apply 115 VAC ± 10%, 400 Hz ± 5% input power to the heater blanket. Continue testing at these parameters until battery electrolyte

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temperature exceeds -10°C (+14°F). Allow the entire battery to stabilize at room temperature for not less than 16 hours. Perform the dielectric strength test of 4.5.27. Heater blanket shall comply with 3.9, 3.10.27, and 3.10.32."

"4.5.34 Heater blanket life test. The battery heater shall be tested for 500 hours under the following operating conditions. Insure the heater blanket is not installed in a battery; if connected to the harness, bypass the two heater thermoswitches.

a. Test at 60 ± 3°C (140°F), ambient humidity and atmospheric pressure.

b. Cycle the heater input power by applying 115 VAC ± 10%, 400 Hz ± 5% input power to the heater blanket for 30 ± 5 seconds every 5 minutes.

c. At the 250th and 499th hours, increase the input voltage to 190 volts, RMS, for not less than 0.1 second.

At the completion of 500 hours, examine the heater for deterioration or damage. Conduct the dielectric strength test of 4.5.27. The heater blanket shall comply with 3.9, 3.10.27, and 3.10.32.

1.3 Modify the Tables of MIL-PRF-81757 as follows:

Table II. Identification of defects. Add the following:

Number	Description	Method of inspection
15	Thermal switches not as specified at room temperature.	Meter
16	Incorrect transducer values.	Meter
17	Date codes missing or incorrect or more than one date code on a battery.	Visual
18	Connector mounted incorrectly	Visual

Table IV. Qualification inspection of components and materials. Add the following:

Part	Examination and tests	Requirement paragraph	Test paragraph
Wiring harness	WIRING HARNESS TEMPERATURE TEST	3.10.31	4.5.32
Wiring harness	HEATER BLANKET OPERATION TEST	3.10.32	4.5.33
Wiring harness	HEATER BLANKET LIFE TEST	3.10.32	4.5.34

Table VII. Group B inspection. Add the following:

Test No.	Examinations and tests	Requirement paragraph	Test paragraph
I-4	Dielectric strength and insulation resistance (wiring harness only)	3.9, 3.10.27	4.5.27

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Table VIII. Group C inspection. Add the following:

Test No.	Examinations and tests	Requirement paragraph	Test paragraph
	Subgroup I		
I-10	Electrolyte resistance (wiring harness only)	3.9, 3.10.3	4.5.3
I-11	Heater blanket operation test	3.10.32	4.5.33
	Subgroup II		
II-8	Wiring harness temperature test	3.10.31	4.5.32

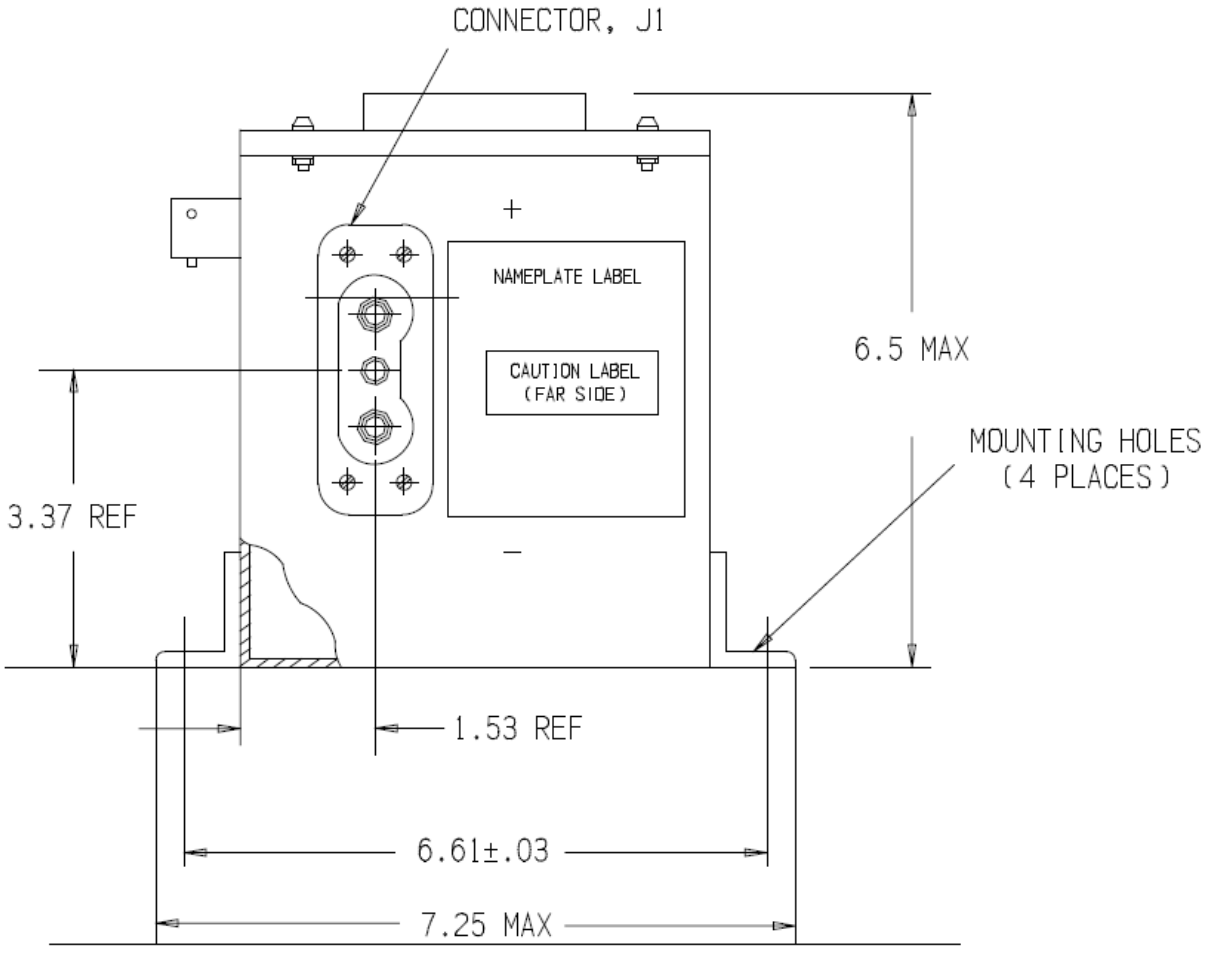


FIGURE 1: Battery Assembly BB-664/A Front View

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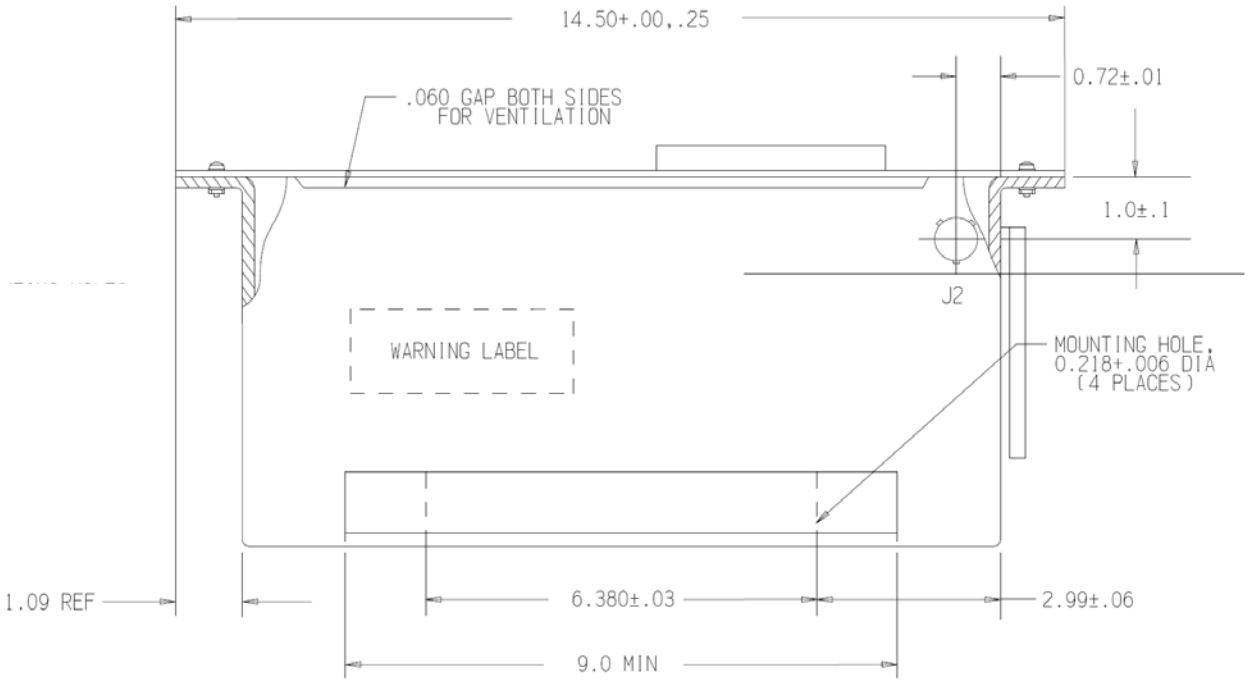


FIGURE 2: Battery Assembly BB-664/A, Side View

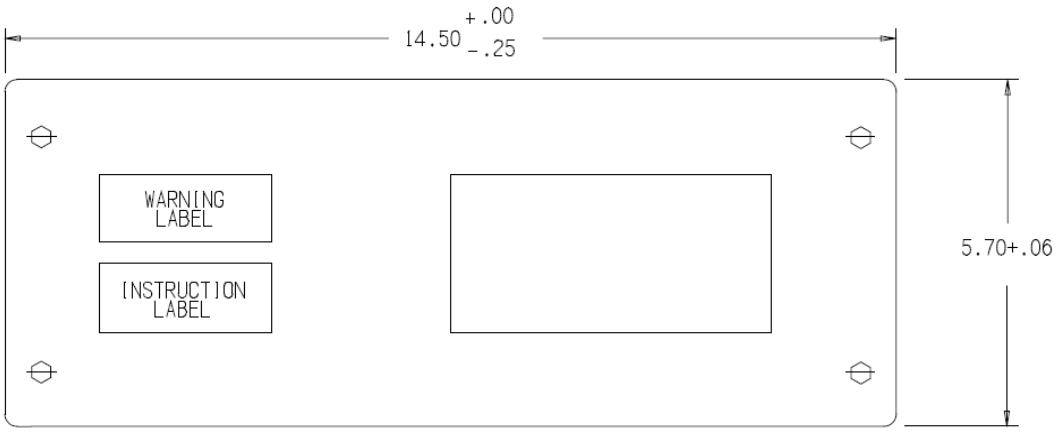


FIGURE 3: Battery Assembly BB-664/A, Top View (Cover)

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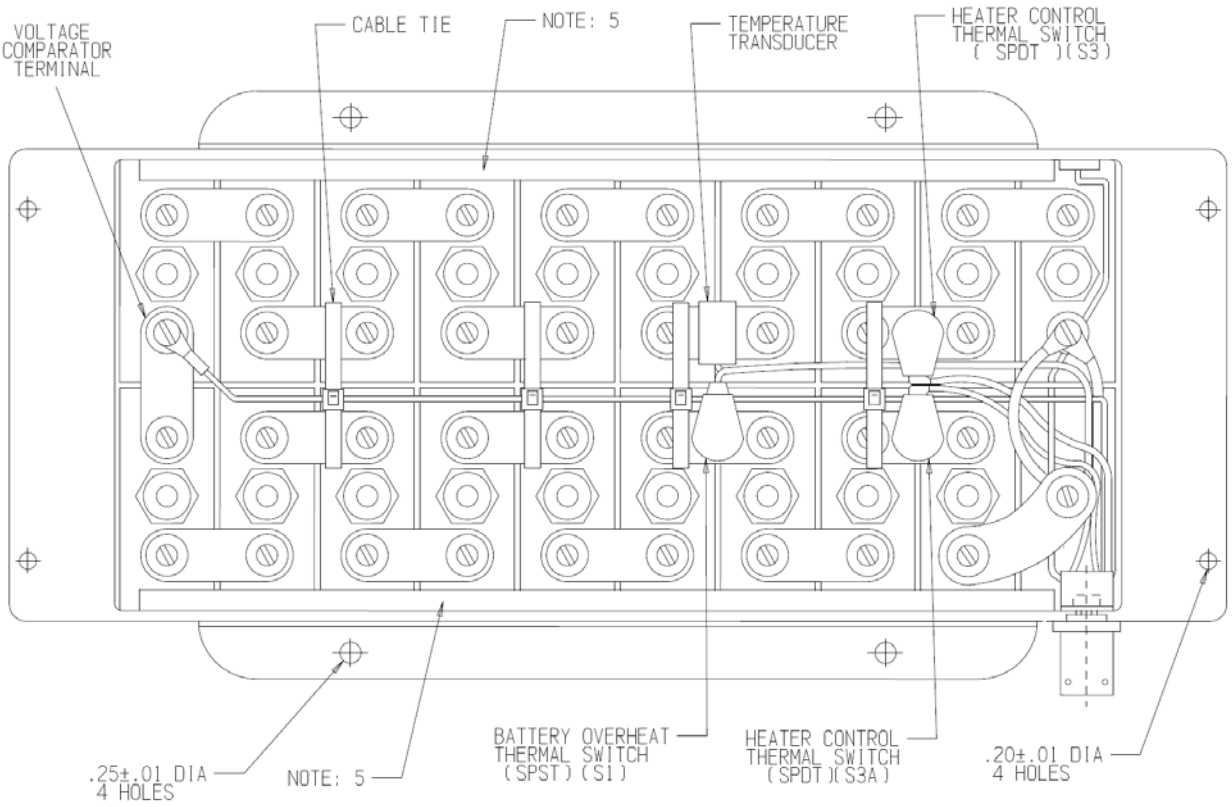


FIGURE 4: Cell Layout Inside BB-664/A, Top View

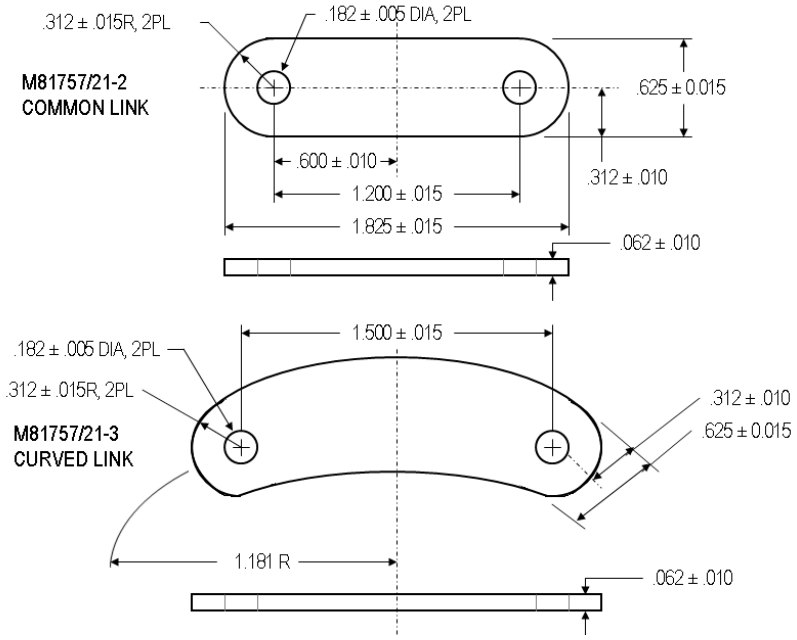


FIGURE 5: Cell Links, BB-664/A

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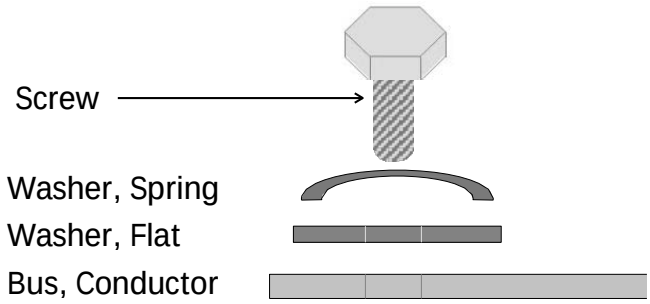


FIGURE 6: Screw/Washer Assembly Detail

FIGURE NOTES:

1. All dimensions are in inches, unless otherwise specified.

2. Battery Container. The body and cover shall be made of the same alkali-resistant material (see 3.7.2 of the general specification as modified herein). The dimensions of the container and the dimensions and locations of the receptacle mounting holes shall conform to Figures 1, 2, & 3.

3. Bottom flange mounting. The battery shall have a bottom flange on each side of the battery as shown in figures 1 & 2.

4. Container venting. The battery shall provide a 0.06” gap between container and cover for ventilation.

5. Cell hold-downs. The cells shall be held in place by nylon bars. The nylon bars shall hold down on the narrow shoulder of the cells along the long inside surface of the battery case. They shall be fastened in a manner which does not cause conductive materials to be exposed to the interior of the battery.

6. Raised area on battery cover. The raised area on the battery cover (Figures, 1, 2, & 3) is to accommodate thermal switches and the temperature transducer, along with mounting hardware. The internal dimensions of this raised area are 2.88 ±0.06 inches by 4.95 ± 0.06 inches. The internal edge of the raised area shall be 2.54 ± 0.06 inches from the J1 connector side of the battery. The raised area shall be centered between the sides of the cover to within 0.12 inches.

7. Hardware. The following hardware is required to complete the cell assembly inside the BB-664/A:

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ITEM	NSN	P/N (CAGEC)	QUANTITY
Wiring Harness	5995-01-315-1364	M81757/23-1 (81349) 0227141-000 (09052)	1
Bus, Conductor Bus, Conductor [1]	6150-01-041-4581 ---	M81757/21-2 (81349) 10371-114 (74025) M81757/21-3 (81349)	15 1
Screw, Hex-head, SS, #8-32 UNC 3A, 7/16”L			38
Washer, Spring Tension	5310-00-118-1977	M81757/1-8A (81349) 16128-26 (74025)	38
Washer, Flat	5310-00-976-4621	162A7819-22 (01002)	38

Notes:
NSN = National Stock Number
P/N = Part Number
CAGEC = Commercial and Government Entity Code
[1] Connects positive termination of cell assembly to MS3509 connector

Note: Potassium carbonate. Previous versions of specifications for the Army Apache battery had requirements and test methods for potassium carbonate. It has been determined that failure to control and limit potassium carbonate in production would result in cells incapable of complying with the performance requirements herein. Users of this document should be aware that there remains a need to limit potassium carbonate levels in order to provide assurance of capability in both qualification and conformance inspections.

CHANGES FROM PREVIOUS ISSUE: The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army – CR

Preparing activity
Army – CR
(Project Number 6140-2016-002)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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